Experiment: Construction and Simulation of a NOR Gate Using MOS Transistors SCIENCE LAB REPORT

GROUP 61 :

Aim

To construct and simulate a NOR gate using MOS transistors, observe its output for different input combinations, and verify its truth table and logical operation

MATERIALS

Two NMOS transistors

Two PMOS transistors

DC power supply ($V_{DD} = 5V$, depending on the MOSFET model)

Input voltage sources A and B

Ground connection

Simulation software such as LTspice, Proteus, or Multisim

Theory

A NOR gate is a digital logic gate whose output is HIGH (1) only when all its inputs are LOW (0). It performs the operation of an OR gate followed by a NOT gate.

The Boolean expression is:

$$Y = \overline{A + B}$$

Here:

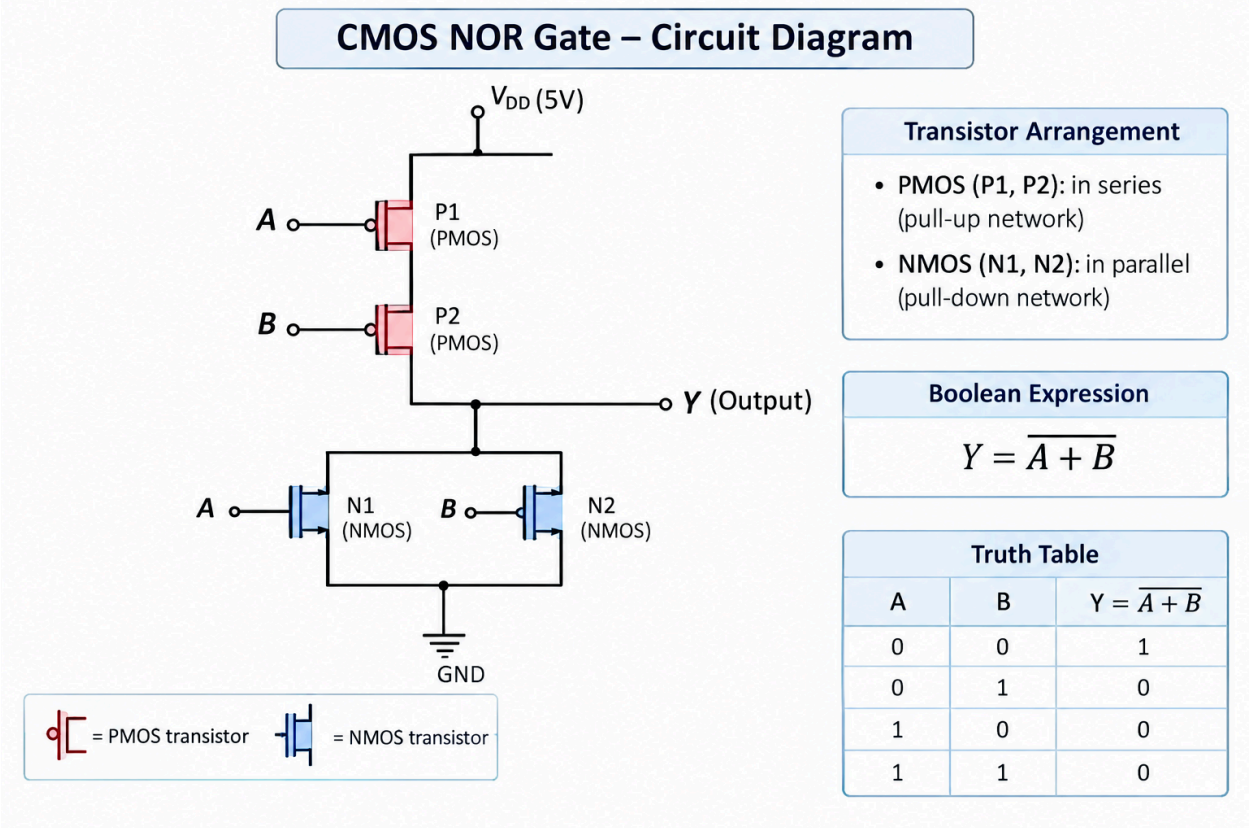
- A and B are the input signals.
- $+$ represents the OR operation.
- The overbar represents the NOT operation.
- Y is the output signal.

MOS transistor implementation

A CMOS NOR gate uses four MOS transistors: two PMOS transistors in series in the pull-up network and two NMOS transistors in parallel in the pull-down network.

- PMOS transistors: Connected in series between V_{DD} and the output. Both must conduct to produce a HIGH output.
- NMOS transistors: Connected in parallel between the output and ground. Either transistor can conduct to pull the output LOW.

Circuit Diagram



The circuit connections are as follows:

- PMOS network: Connect the two PMOS transistors in series between V_{DD} and output Y.
- NMOS network: Connect the two NMOS transistors in parallel between output Y and ground.
- Connect input A to the gate of one PMOS and one NMOS.

- Connect input *B* to the gate of the other PMOS and NMOS.
- Take the output *Y* from the junction of the pull-up and pull-down networks.

STIMULATION

https://www.tinkercad.com/things/eJYWhnaz1MH-nor-gate-using-cmos/editel?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard&sharecode=Ffzm-7cpIqg7vfIjqTH7fd6qpUa7tsylg15qydJUuLU

Truth Table

Input A	Input B	Output Y = (A + B)’
0	0	1
0	1	0
1	0	0
1	1	0

Here, 0 represents LOW voltage and 1 represents HIGH voltage.

Working and Logical Operation

00 : Both inputs LOW

Both PMOS transistors conduct, while both NMOS transistors remain OFF. The output is HIGH (1).

01 : A LOW, B HIGH

One PMOS turns OFF and one NMOS turns ON. The output is LOW (0).

10 : A HIGH, B LOW

One PMOS turns OFF and one NMOS turns ON. The output is LOW (0).

11 : Both inputs HIGH

Both NMOS transistors conduct, pulling the output to ground. The output is LOW (0).

PROCEDURE

1. Open LTspice, Proteus, or another suitable circuit simulation tool.
2. Place two PMOS and two NMOS transistor models in the schematic.
3. Connect the PMOS transistors in series and the NMOS transistors in parallel.
4. Connect the input sources A and B to the corresponding transistor gates.
5. Connect the supply voltage and ground.
6. Apply all four input combinations: 00, 01, 10, and 11.
7. Run the simulation and measure the output voltage Y.
8. Compare the observed outputs with the theoretical truth table.

RESULTS

A NOR gate was constructed and simulated using MOS transistors. Its output behavior was observed for all possible input combinations, and the truth table was verified. The NOR gate performs the Boolean operation $Y = \overline{A + B}$.

Inference

The output of the NOR gate is HIGH only when both inputs are LOW. If either input or both inputs become HIGH, the output becomes LOW. Therefore, the simulated circuit verifies the NOR logical operation.